

Question No. 4

(a) PMOS:- $V_{SG} = 5V$ $V_{SD} = 2.5V$ $V_{TH} = -1V$ $\frac{W}{L} = ?$

NMOS:- $V_{GS} = 2.5V$ $V_{DS} = 2.5V$ $V_{TO} = 1V$ $\frac{W}{L} = 10$

Both in saturation region; PMOS Linear Region.

$$\left(\frac{0.6}{2}\right)(10)[2.5-1]^2 = \left(\frac{0.2}{2}\right)\left(\frac{W}{L}\right)_p [4 - 1 \cdot 2.5] 2.5$$

$$\boxed{\left(\frac{W}{L}\right)_p \approx 5}$$

No Marks if CMOS inverter eqns. are used **6 Marks**

(b) $\left(\frac{W}{L}\right)$ is equal for NMOS and PMOS 10.

$$V_{NH, low} = V_{IL} - V_{OL} \quad (\text{here } V_{OL} \neq 0)$$

$$V_{OL}:- (V_{in} = V_{OH} = 5V)$$

NMOS ($V_{DS} \approx 0$ Linear) PMOS (saturation region)

$$\left(\frac{0.2}{2}\right)(10)[5-1]^2 = 0.6 \times 10 \left[4 - \frac{V_{out}}{2}\right] V_{out}$$

$$\boxed{V_{OL} = 0.734V} \quad \text{3 Marks}$$

$V_{IL}:- (V_{in} \text{ Low}) (V_{out} \text{ High})$ exact value not known

NMOS Saturation and PMOS Linear

$$(V_{out} + 3)(5 - V_{out}) = 3(V_{in} - 1)^2 \rightarrow (1)$$

$$\frac{dV_{out}}{dV_{in}} = -1 \quad \text{Limiting condition on Eq-1}$$

(1)

Q.N0.4

$$V_{out} = 3V_{in} - 2 \rightarrow (2) \text{ using Eq. 2 in Eq. 1}$$

$$V_{IL} = 2.22V, V_{out} = 4.66V \quad \text{3 Marks}$$

$$\text{Noise Margin Low} = 1.48V \quad \text{2 Marks}$$

c) Static power dissipation:-

$$\begin{aligned} & \left(\frac{0.2}{2}\right)(10) [5-1]^2 \times V_{DD} \text{ mW} \\ & = 1 \times 16 \times 5 \text{ mW} = 80 \text{ mW} \quad \text{4 Marks} \end{aligned}$$

d) Worst case propagation delay:-

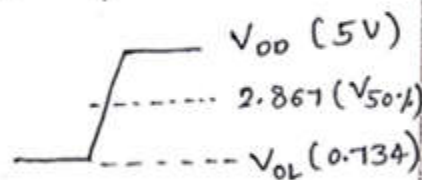
τ_{PLH} (Low to High propagation delay)

PMOS current charges the load capacitor 200 fF .

$V_{out} = V_{OH}$ PMOS saturation

$$I_{PMOS} = \left(\frac{0.2}{2}\right) \times 10 \times (5-1)^2 \text{ mA}$$

$$= 16 \text{ mA}$$



$V_{out} = V_{50\%}$ PMOS Linear $V_{SG} = 5V$; $V_{SD} = 2.133V$

$$I_{PMOS} = 0.2 \times 10 \left[4 - \frac{2.133}{2} \right] 2.133 \text{ mA} = 12.51 \text{ mA}$$

$$I_{average} = 14.25 \text{ mA}$$

3 Marks

$$\tau_{PLH} = 29.93 \text{ ps}$$

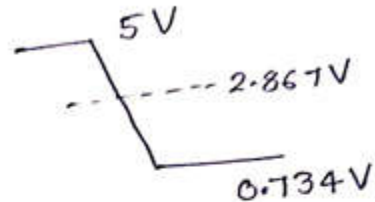
Worst case
2 Marks delay.

Q.No.4

T_{PHL} (High to Low propagation Delay):-

Current through Cap. = $I_{NMOS} - I_{PMOS}$
(Discharge current)

$V_{out} = 5V$:- NMOS sat.
PMOS Linear.



$$I_{NMOS} = \left[\frac{0.6 \times 10^{-6}}{2} \right] (5-1)^2 = 48 \text{ mA}$$

$$I_{PMOS} \approx 0 \quad (V_{SD} = 0V)$$

$V_{out} = 2.867V$ NMOS Linear; PMOS Linear.

$$I_{NMOS} = 44.148 \text{ mA} \quad I_{PMOS} = 12.51 \text{ mA}$$

$$I_{Cap.} = 31.63 \text{ mA}$$

$$\text{Average current} = \boxed{39.81 \text{ mA}}$$

$$\boxed{T_{PHL} = 10.71 \text{ ps}}$$

2+1 Marks

③ Dynamic power dissipation:- $d = 0.5$

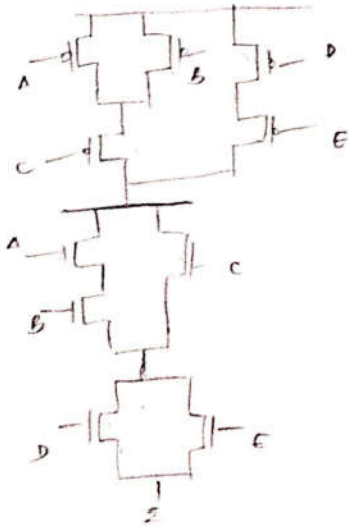
$$f_{max} = \frac{1}{T(T_{PHL} + T_{PLH})} = 3.515 \text{ GHz}$$

$$P_{dyn.} = T \times 0.5 \times 5 \times 4.266 \times 3.515 \times 200 \times 10^{-6}$$

$$= \boxed{52.48 \text{ mW}}$$

4 Marks

①
②



①
A - C - E - D - B
A - B - D - E - C
C - E - D - B - A
B - D - E - C - A
B - A - C - E - D

③

③

$$\frac{1}{S} = \frac{1}{4.54} \quad \cdot \quad \frac{1}{\beta} = \frac{1}{2} \quad \cdot \quad \alpha = \frac{S}{\beta} = 2.27$$

i velocity $\rightarrow$ constant

ii $I_D \rightarrow \frac{1}{\beta} = 8 \mu A$

⑤

iii $P_{density} = 61.83 \text{ mW/cm}^2$

iv. gate delay = 0.22

v $EDP = \frac{1}{(\Delta\beta)^2} = 0.062$

②

$t_{inv} = 0.7 \text{ ns}$

$$P_{L1} = 0.7 \times RC \times L^2$$

$$= 35 \text{ nA}$$

$$P_{L2} = 8.75$$

critical path = $5 + 10 + 35 + 10 + 8.75 + 10$
 $= 78.75$

③

$$m_{LI} = \frac{L}{\sqrt{\frac{RC}{2 \times R_i C_i}}} = 10 \sqrt{\frac{0.5 \times 1 \times 10^{-9}}{2 \times 1 \times 10^{-9}}} = 5$$

$$m_{12} = 5 \times \sqrt{\frac{0.5 \times 1}{2 \times 1}} = 2.053$$

$$\textcircled{b} 0.7 \times 5 \times \left(\frac{L}{5}\right)^2 + 4 \times 0.7 \text{ ns} \Rightarrow P_1$$

$$= 17 + 2.8 \text{ ns} = 19.8 \text{ ns}$$

For 2.

$$0.7 \times 2 \times \left(\frac{5}{2}\right)^2 + 0.7 \text{ ns} = 5.075 \text{ ns}$$

For 3

$$0.7 \times 3 \times \left(\frac{5}{3}\right)^2 + 2 \times 0.7 = 4.316 \text{ ns}$$

$$\text{Delay} = 5 + 3 \times 10 + 19.8 + 4.316$$

$$= 49.116$$

① After restructuring

$$= 45.116 \text{ ns}$$

$$\textcircled{3} \textcircled{a} \beta_{\text{opt}} = \sqrt{\frac{45}{15}} = 1.7$$

$$\textcircled{b} C_{in} = 2.7 \times C_{ox} \times 0.45 \times 0.25 = 3.03 \text{ pf}$$

①

	g	P
NOT gate	1	1
2 input NOR	1.63	2.
2 input NAND	1.37	2
3 input NOR	2.26	3

$$G = 5.02$$

$$b = 2 \times 3 = 6$$

$$H = 5$$

$$F = 150.7$$

$$D^{\wedge} = 22.63 \text{ d.u.}$$

$$N^{\wedge} = 3.6 \approx 4$$

3 marks for rest of the

4

$$\textcircled{5} C_{\text{logic}} = 10 \times 10^6 \times 12 \times 0.09 \times 2 \times 10^{-15}$$

$$= 21.6 \text{ nF}$$

$$C_{\text{mem}} = 20 \times 10^6 \times 4 \times 0.09 \times 2 \times 10^{-15}$$

$$= 14.4 \text{ nF}$$

$$P = 2(21.6 \times 0.5 + 14.4 \times 0.25) \times 10^{-9} \times (1.8)^2 \times f$$

$$\textcircled{a} f = 0.428 \text{ MHz or } 428.6 \text{ kHz}$$

2

No Marks when the answer is orders of magnitude away from the final answer

$$\textcircled{b} P_{\text{static}} = [5 \times 10^6 \times 12 \times 0.09 + 10 \times 10^6 \times 4 \times 0.09] \times 22 \times 10^9 \times 1.8$$

$$= 356.4 \text{ mW}$$

2

$$\textcircled{c} P_{\text{ghf}} = 2[(6.5 \times 10^3 + 0.25 \times 10^2) \times 0.2 \times 10^3 \times 0.5 \times 10^{-12} \times 1.8 \times 200 \times 10^6]$$

$$= 37.8 \text{ uW}$$

3